# Patient Flow Simulator — Decision Report

Type 1 Emergency Department

## Executive summary

- Modelled demand: 325 attendances/day (1,952 patients over the simulated period).
- 4-hour performance: 79%  (95% CI 70-80% across 24 Monte-Carlo runs) — 417 breaches.
- Mean time in department: 2.8 h (95% CI 2.8-3.4 h); 95th percentile 8.2 h.
- Admitted: 16% of attendances.

## Bottleneck diagnosis

Biggest delay: SDEC — patients wait 111 min on average (681 patient-hours lost over the run).

Recommendation: sdec doctor is busy (75% utilisation) and likely drives this queue; trial extra capacity in Scenario comparison.

## What-if scenarios (ranked)

| Scenario | 4-hr perf. | vs baseline | Mean LOS |
|---|---|---|---|
| Baseline | 75% | — | 3.0 h |
| +3 sdec doctor | 85% | **+10 pts** | 2.5 h |
| +8 inpatient bed | 75% | **+0 pts** | 3.0 h |
| Winter +15% demand | 70% | **-5 pts** | 3.5 h |

For demonstration — synthetic data seeded from the beta config.json. Validate against local data before operational use.

# Patient Flow Simulator

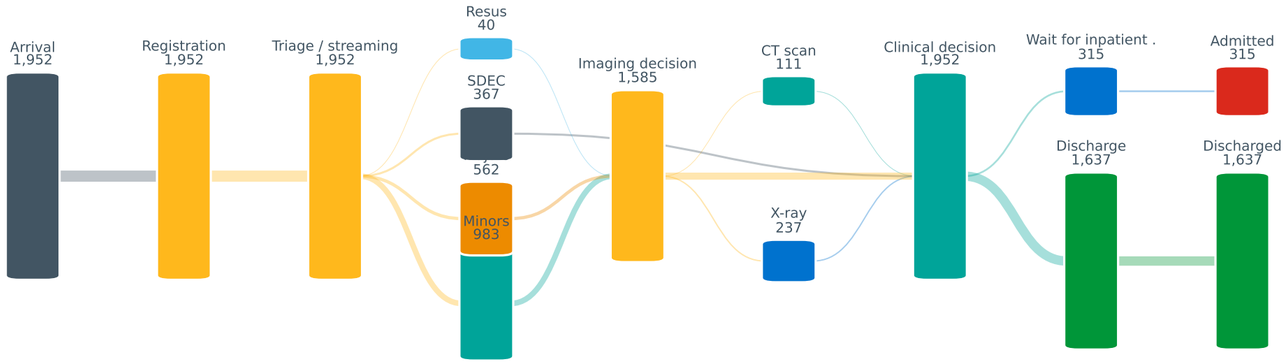
Discrete-event simulation for emergency & urgent care planning

**Type 1 Emergency Department**

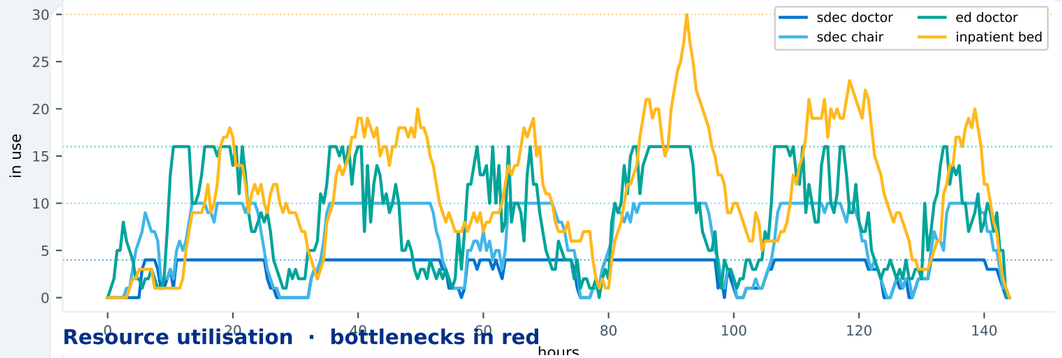
325 attendances/day · 6-day run · 24 Monte-Carlo replications

| 4-HOUR PERFORMANCE | MEAN TIME IN DEPT | 95TH PERCENTILE LOS | ADMITTED | DAILY ATTENDANCES |
|---|---|---|---|---|
| **79%** | **2.8h** | **8.2h** | **16%** | **325** |
| 95% CI 70-80% | 95% CI 2.8-3.4h | longest typical wait | 7.2h mean LOS | 1,952 simulated |

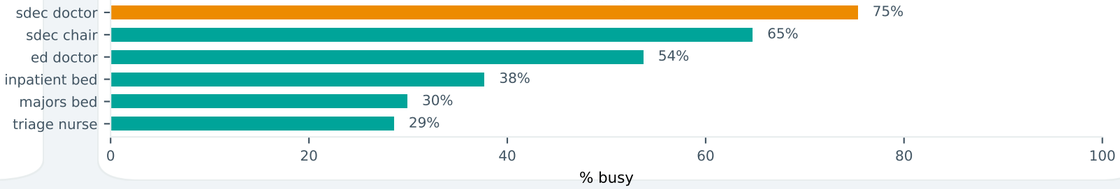
## Patient flow · arrival to disposition



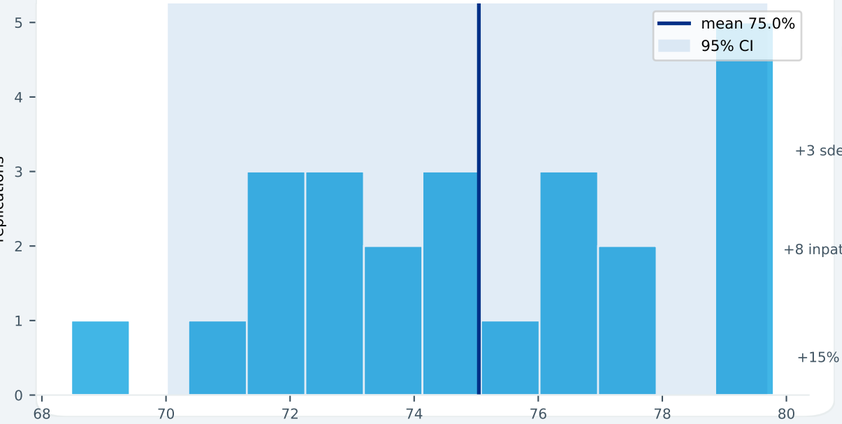

## Resource occupancy vs capacity (dotted)



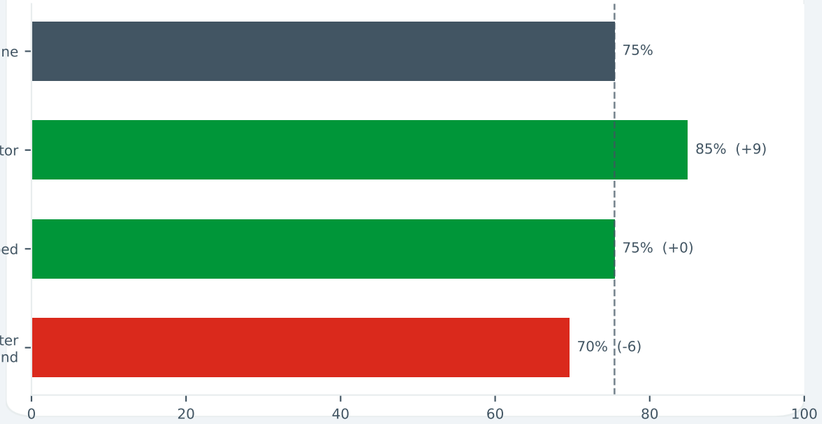

## Resource utilisation · bottlenecks in red




## Monte-Carlo: 4-hour performance variability




## Scenario comparison · 4-hour performance




## Where patients wait · mean queue time

| STAGE | WAIT | SEEN |
|---|---|---|
| SDEC | **111m** | 367 |
| CT scan | **8m** | 111 |
| Resus | **4m** | 40 |
| Minors | **3m** | 983 |
| Majors | **3m** | 562 |
| X-ray | **1m** | 237 |